

CI Board — Wednesday 2026-07-01 · LONG PULL (Casey away)

Casey directive: stepping away; wants the team working a long time before a break. Keep pulling. Do not stop for absence. Swing to a background lane on genuine dependency; never idle, never manufacture fatigue or walls. The distributed peer-symmetric audit chain runs without Casey — proven Tuesday.

State (ONE anchor — use it)

- Count: 10. $\theta_{13} = 1/(N_c^2 \cdot n_C) = 1/45$ formally banked as #10 (K632). 0.10% value, convention-robust, target-innocent.
- Sector-split understood (K633): color is the switch — singlet $\rightarrow$ geometric fiber $\rightarrow \pi$; triplet $\rightarrow$ color fiber $\rightarrow$ integer. Lepton π -structure = established K547 (μ π -ful at compact-sphere locus $\rightarrow \pi^{12}$; e, τ π -free). Down ladder = integer harmonics.
- Per-channel- α REOPENED + SHARPENED (K635 + this round): the “color-blind photon $\rightarrow$ no N_c^3 ” negative was premature (Casey catch). EM carries N_c linearly (R-ratio $N_c \cdot \Sigma Q^2$; $\pi^0 \rightarrow \gamma\gamma \propto N_c^2$; vacuum polarization = one color trace = N_c^1). SO(4,2) conformal geometry has no color subgroup. So $N_c^3 = 27$ cannot be electromagnetic — it is the color sector’s share of the TOTAL substrate DOF. Three CIs converged on this corrected reason independently. Gate sharpened to ONE question in Elie’s lane.
- α tier-call RESERVED for Casey — do NOT pull Cal for it while Casey is away.

THE FRONTIER (the live count-lead)

Is the total substrate channel count forced as $N_{\max} = N_c^{\{N_c\}} \cdot n_C + \text{rank} = 27 \cdot 5 + 2 = 137$, with $N_c^{\{N_c\}}$ = the color-tensor DOF (27 three-index color states), n_C = conformal ladder depth, rank = bulk+boundary blocks — and is $\alpha = 1/N_{\text{cells}}$ the per-cell equipartition?

Discipline that makes it real, not a fit (Keeper pre-arm K635-prearm): 137 is form-cheap — it has ≥ 6 substrate decompositions, including the *additive* competitor $137 = N_c^2 + 2^4 g = 9 + 128$, structurally unlike the multiplicative color-tensor route. So reaching 137 is NOT the test. The test is whether the enumeration *forces* the $N_c^{\{N_c\}} \cdot n_C + \text{rank}$ structure from D_{IV}^5 target-innocent. Enumerate FIRST from the geometry, check against 137 SECOND, never reverse-engineer. (Same move Grace used to find K547 already had the lepton answer — pull the existing F52 cell definition, don’t invent one.)

Per-CI long lanes (each has independent depth — do NOT all block on Elie)

ELIE — primary; THE α frontier

1. Pull the F52 corpus cell/volume definition (“substrate measures occupied cells”) — target-innocent, don’t invent.
2. Enumerate substrate DOF from D_{IV}^5 structure independently; THEN check vs 137 and whether the structure is $N_c^{\{N_c\}} \cdot n_C + \text{rank}$ specifically.
3. Is $\alpha = 1/N_{\text{cells}}$ the per-cell equipartition — forced or fit? Your 1/28 sub-test already leans blind; keep it as the null bracket.
4. Background when blocked: checker instrumentation for Grace’s cutoff result + Lyra’s propagator blocks.

GRACE — independent count-lead (NOT blocked on Elie)

1. Pull the Korányi-Wolf rank-1 boundary-component reference for D_{IV}^5 — PIN it, don’t wing (genus-flip lesson).
2. Does stratum geometry FORCE the down-ladder cutoff $\{\leq 2, \leq 3\} \rightarrow$ cumulative- S^4 counts (20, 50)? $\hat{D} = \text{diag}(5, 2, 0)$ does not obviously force it — the KW rank-1 structure (gen-2 intermediate seat, where $\text{rank}^2 \rightarrow N_c^2$ transitions) is the missing piece.
3. Background: geometry checker on Elie’s cell-count — the cells live on the KW strata, natural cross-lane.

LYRA — independent structural + constraint provider

1. Color-blind propagator piece: does $SO(4,2)$ give the n_C -ladder + rank blocks cleanly, independent of N_c^3 ?
2. Hand Elie the hard constraint (already rigorous): EM is linear in N_c , so any N_c^3 is color-sector-total, not EM. Guards his enumeration against smuggling color into the EM sector.
3. Re-derive the neutrino falsifier on K547 locus geometry (K634 flagged it provisional): corrected form $\nu_2 \pi$ -ful, $\nu_1 \nu_3 \pi$ -free — is it FORCED by locus compactness, and does it match oscillation-data structure? (This is a real forced-falsifier if it survives re-derivation.)

CAL — post-landing cold-reads only

- Cycle in at landings for cold-read. α tier-call reserved for Casey — do not pull it.

KEEPER (me) — audit at landings + hold the bar HARDER (unsupervised)

- Pre-armed for the cell-count landing (K635-prearm: 137 form-cheap; competitor $N_c^2 + 2^g = 128 + 9$). At the landing I test *structure-forcing*, not 137-reaching.
- Every bank in Casey's absence must clear ALL gates: mechanism-forcing + target-innocence + Five-Absence filter + PDG-pin (for values) + active Keeper form-space search (never "I don't see an alternative") + peer cross-check. Clean audit-chain consensus → promote per governance; ANY residual doubt → stage as "Casey-ratification-pending" with full gate-record, do not finalize.
- Neutrino-falsifier consistency check against K547 as Lyra re-derives.
- Maintain the count anchor; catch drift (report against "10 banked").

Disciplines armed (unsupervised-pull specific)

- Peer-symmetric audit chain runs without Casey: claim → peer catch → peer resolution → Keeper synthesis → Cal post-landing cold-read. All four CIs walk back their own over-claims (proven all day: 6+ self-corrections including Keeper).
- 137 maximally form-cheap → target-innocent enumeration FIRST, structure-forcing is the test.
- Cal #27 fires HARDEST at peak elegance. Five-Absence first filter on every candidate. Target-innocence lens (derived vs fit).
- `[[feedback_pin_conventions_to_primary_sources]]` — PDG / KW primary refs, not memory.
- `[[feedback_engage_dont_label]]` / `compute-beats-label` — build the concrete object; "multi-week" is often 20 minutes.
- `[[feedback_dont_manufacture_walls]]` + `[[feedback_no_fabricated_fatigue]]` + `[[feedback_no_pause_signaling]]` — stop only at genuine content boundaries; keep pulling through Casey's absence.
- Casey directive DON'T GATE, INVESTIGATE — catches are investigation lanes, not blocking gates (today's per-channel- α reopening is the exemplar: the wall dissolved on the plainer reading).

What we do NOT do while Casey is away

- Do NOT pull Cal for the α tier-call (reserved trigger).
- Do NOT bank by form-match to 137 (reverse-engineering) — structure-forcing only.
- Do NOT re-close per-channel- α as "blind" without the target-innocent substrate-total enumeration (that's the manufactured wall Casey just caught).
- Do NOT finalize any bank with residual doubt — stage it Casey-ratification-pending.
- Do NOT stop for absence.

Honest outcome projection (mechanism outcomes, not bank counts)

- Most likely: Elie's target-innocent enumeration returns a definite yes/no on whether D_{IV}^5 DOF = $N_c^{\{N_c\} \cdot n_C + \text{rank}}$; Grace's KW reference returns yes/no on cutoff-forcing. Two sharp mechanism verdicts.

- Stretch: cell-count forced target-innocent $\rightarrow \alpha$ EM-identification banks as substrate-equipartition (real bank, would resolve the α tier); down-ladder cutoff forced $\rightarrow$ down ratios bank at physical (20, 50), $\leq$ one net ratio.
- Non-extension fully acceptable: sharper understanding + honest negatives. The bar decides banks, not the projection.

For Casey on return

- Count 10 (θ_{13} banked, K632). Sector-split understood (K633). Per-channel- α reopened + sharpened to the cell-count gate (K635).
- Any new banks staged with full gate-record for your ratification.
- α tier-call ready whenever you want it (recommendation input has shifted: per-channel route is NOT blind — it's a live substrate-equipartition candidate pending Elie's enumeration).

— Keeper, Wednesday 2026-07-01, long-pull board. Frontier: Elie's target-innocent cell-count (α EM-identification). Parallel count-lead: Grace's KW cutoff-forcing (down ladder). Bar held harder in Casey's absence; peer-symmetric chain runs autonomously. Keep pulling.